

RMO(INDIA) – 1999

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Prove that the inradius of a right-angled with integer sides is an integer.
 2. Find the number of positive integers which divide 10^{999} but not 10^{998} .
 3. Let ABCD be a square and M, N points on sides AB, BC respectively, such that $m\angle MDN = 45^\circ$. If R is the mid point of MN show that $RP = PQ$ where P, Q are points of intersection of AC with lines MD, ND.
 4. If p, q are roots of the cubic equation $x^3 - 3px^2 + 3q^2x - r^3 = 0$. Show that $p = q = r$.
 5. If a, b, c are sides of a triangle, prove the inequality: $\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3$.
 6. Find all solutions in integers m, n of the equation: $(m-n)^2 = \frac{4mn}{m+n-1}$.
 7. Find the number of quadratic polynomials, $ax^2 + bx + c$, which satisfy the following conditions:
 - (a) a, b, c are distinct.
 - (b) $a, b, c \in \{1, 2, 3, \dots, 1999\}$.
 - (c) $x+1$ divides $ax^2 + bx + c$.

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